The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Zepto-Section-Generating Substrate Yocto-Section Dynamics
Collapse to an Observer-Relevant Yocto-Section Minimizer-Image Core

Abstract

The current primitive-reservoir observer-reduction frontier already moved the live burden below the current atto-section-generating substrate zepto-section dynamics package itself: the active theorem derived the current zepto-section package $\mathfrak{B}^{\text{zeptosec}}$ canonically from a deeper zepto-section-generating substrate yocto-section dynamics package $\mathfrak{C}^{\text{yoctosec}}$. Under the recorded safeguard against recursive depth drift, however, the next honest move at that level is not another same-output deeper-origin relay that leaves $\mathfrak{C}^{\text{yoctosec}}$ unchanged. The sharper benchmark-facing question is whether the full recorded yocto-section package still carries benchmark-neutral ambient freedom once the induced zepto-section package is treated as fixed.

The present note proves that it does. It isolates the *observer-relevant yocto-section minimizer-image core*: the minimizer-image compact body over the recorded zepto-section body, the restricted projection back to that body, the restricted yocto-section functional on that image, the exact-shell image of the recorded zepto-section shell, and benchmark faithfulness. The full exact yocto-section shell is then shown to be forced by that image core, and ambient yocto-section state-space directions outside the minimizer image do not add independent benchmark-facing freedom once the induced zepto-section package is held fixed.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore narrower than another deeper relay and sharper than the full yocto-section package: the live burden now collapses to a smaller observer-relevant yocto-section minimizer-image core. What remains open is to derive or justify that smaller core from still more primitive substrate structure, or else prove a still sharper collapse strictly inside it.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again reached a sharper burden than another same-output relay. The current active-primary theorem already removed the current zepto-section package $\mathfrak{B}^{\text{zeptosec}}$ as the primitive stopping point by showing that, once a benchmark-faithful yocto-section package $\mathfrak{C}^{\text{yoctosec}}$ is fixed, the recorded zepto-section body, functional, and exact shell are its canonical fiberwise effective output. That theorem matters precisely because it blocks a familiar form of recursive depth drift: depth alone no longer counts once the live package datum is unchanged.

The next honest move is therefore not to relay the same yocto-section package through another deeper origin while preserving that package and its same downstream outputs. It is to ask whether the present yocto-section package itself still contains benchmark-neutral ambient freedom that

can be removed without changing the induced zepto-section package or the downstream chain depending on it. The present note takes that sharper internal-collapse route. It shows that the full ambient yocto-section state space and off-image body directions are not independently live once one retains only the minimizer image over the recorded zepto-section body together with the restricted functional and exact-shell image it already determines.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, and it does not reopen the already-screened current zepto-section package, the current observer-relevant zepto-section minimizer-image core, or the larger reconstructed downstream presentations as if they were again the live burden. Its narrower benchmark-facing role is to isolate the smaller current datum inside the yocto-section package itself.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded zepto-section downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active line already fixes the following downstream data and consequences:*

1. *the recorded atto-section target data*

$$\mathcal{A}_\sigma^{\mathrm{attosec}}, \qquad \mathcal{B}_\sigma^{\mathrm{attosec}}, \qquad \mathscr{A}_\sigma^{\mathrm{attosec}}, \qquad \mathcal{Z}_\sigma^{\mathrm{attosec}};$$

2. *the current observer-relevant zepto-section minimizer-section core* $\mathfrak{B}^{\mathrm{zep,sec}}$*, the current observer-relevant zepto-section minimizer-image core* $\mathfrak{B}^{\mathrm{zep,img}}$*, the current atto-section package* $\mathfrak{A}^{\mathrm{attosec}}$*, the current femto-section package* $\mathfrak{Z}^{\mathrm{femtosec}}$*, the current pico-section package* $\mathfrak{Y}^{\mathrm{picosec}}$*, the current observer-relevant pico-section minimizer-section core* $\mathfrak{Y}^{\mathrm{pic,sec}}$*, the current nano-section package* $\mathfrak{X}^{\mathrm{nanosec}}$*, the current micro-section package* $\mathfrak{W}^{\mathrm{microsec}}$*, and the current sub-section package* $\mathfrak{V}^{\mathrm{subsec}}$*;*

3. *the previously recorded fact that, once the current zepto-section package is available, the current observer-relevant zepto-section minimizer-section core, the current observer-relevant zepto-section minimizer-image core, the current atto-section package, the current femto-section package, the current pico-section package, the current observer-relevant pico-section minimizer-section core, the current nano-section package, the current micro-section package, the current sub-section package, and the downstream shell-transport consequences used on the active line follow unchanged.*

Definition 3 (Recorded femto-section-generating substrate atto-section dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The recorded current atto-section dynamics package* $\mathfrak{A}^{\mathrm{attosec}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean atto-section state space* $\mathcal{A}_\sigma^{\mathrm{attosec}}$*, a compact convex atto-section body*

$$\mathcal{B}_\sigma^{\mathrm{attosec}} \subset \mathcal{A}_\sigma^{\mathrm{attosec}}$$

with nonempty interior, and an affine surjection

$$\Pi_\sigma^{\mathrm{attosec}} : \mathcal{A}_\sigma^{\mathrm{attosec}} \to \mathcal{R}_\sigma^{\mathrm{femtosec}}$$

satisfying

$$\Pi_\sigma^{\mathrm{attosec}}(\mathcal{B}_\sigma^{\mathrm{attosec}}) = \mathcal{B}_\sigma^{\mathrm{femtosec}}.$$

For each $t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}$*, write*

$$\mathcal{F}_\sigma^{\mathrm{attosec}}(t) \equiv \left(\Pi_\sigma^{\mathrm{attosec}}\right)^{-1}(t) \cap \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth atto-section functional*

$$\mathscr{A}_\sigma^{\mathrm{attosec}} : \mathcal{A}_\sigma^{\mathrm{attosec}} \to [0, \infty)$$

such that, for every $t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}$*, the restriction of* $\mathscr{A}_\sigma^{\mathrm{attosec}}$ *to* $\mathcal{F}_\sigma^{\mathrm{attosec}}(t)$ *is strictly convex and attains a unique minimizer. The resulting minimizer depends smoothly on* t*; write it as*

$$\mathfrak{a}_\sigma^{\mathrm{attosec}} : \mathcal{B}_\sigma^{\mathrm{femtosec}} \to \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

These data satisfy

$$\Pi_\sigma^{\mathrm{attosec}} \circ \mathfrak{a}_\sigma^{\mathrm{attosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{femtosec}}},$$

and the effective minimum value recovers the recorded femto-section functional:

$$\mathscr{F}_\sigma^{\mathrm{femtosec}}(t) = \mathscr{A}_\sigma^{\mathrm{attosec}}\big(\mathfrak{a}_\sigma^{\mathrm{attosec}}(t)\big) = \min_{v \in \mathcal{F}_\sigma^{\mathrm{attosec}}(t)} \mathscr{A}_\sigma^{\mathrm{attosec}}(v) \qquad \text{for every } t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}.$$

3. *A closed embedded exact atto-section shell*

$$\mathcal{Z}_\sigma^{\mathrm{attosec}} \subset \mathcal{B}_\sigma^{\mathrm{attosec}}$$

satisfying

$$\mathcal{Z}_\sigma^{\mathrm{attosec}} = \left\{ v \in \mathcal{B}_\sigma^{\mathrm{attosec}} : \begin{array}{l} \Pi_\sigma^{\mathrm{attosec}}(v) \in \mathcal{Z}_\sigma^{\mathrm{femtosec}}, \\ \mathscr{A}_\sigma^{\mathrm{attosec}}(v) = \min\limits_{w \in \mathcal{F}_\sigma^{\mathrm{attosec}}(\Pi_\sigma^{\mathrm{attosec}}(v))} \mathscr{A}_\sigma^{\mathrm{attosec}}(w) \end{array} \right\},$$

and

$$\Pi_\sigma^{\mathrm{attosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{attosec}}} : \mathcal{Z}_\sigma^{\mathrm{attosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{femtosec}}$$

is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as the fixed benchmark-relevant atto-section package used on the active line.*

Definition 4 (Recorded atto-section-generating substrate zepto-section dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The recorded current zepto-section dynamics package* $\mathfrak{B}^{\mathrm{zeptosec}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean zepto-section state space* $\mathcal{U}_\sigma^{\mathrm{zeptosec}}$*, a compact convex zepto-section body*

$$\mathcal{B}_\sigma^{\mathrm{zeptosec}} \subset \mathcal{U}_\sigma^{\mathrm{zeptosec}}$$

with nonempty interior, and an affine surjection

$$\Pi_\sigma^{\mathrm{zeptosec}} : \mathcal{U}_\sigma^{\mathrm{zeptosec}} \to \mathcal{A}_\sigma^{\mathrm{attosec}}$$

satisfying

$$\Pi_\sigma^{\mathrm{zeptosec}}\big(\mathcal{B}_\sigma^{\mathrm{zeptosec}}\big) = \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

For each $v \in \mathcal{B}_\sigma^{\mathrm{attosec}}$*, write*

$$\mathcal{F}_\sigma^{\mathrm{zeptosec}}(v) \equiv \big(\Pi_\sigma^{\mathrm{zeptosec}}\big)^{-1}(v) \cap \mathcal{B}_\sigma^{\mathrm{zeptosec}}.$$

Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth zepto-section functional*

$$\mathscr{U}_\sigma^{\mathrm{zeptosec}} : \mathcal{U}_\sigma^{\mathrm{zeptosec}} \to [0, \infty)$$

such that, for every $v \in \mathcal{B}_\sigma^{\mathrm{attosec}}$*, the restriction of* $\mathscr{U}_\sigma^{\mathrm{zeptosec}}$ *to* $\mathcal{F}_\sigma^{\mathrm{zeptosec}}(v)$ *is strictly convex*

and attains a unique minimizer. The resulting minimizer depends smoothly on v; write it as

$$\mathfrak{z}_\sigma^{\mathrm{zeptosec}} : \mathcal{B}_\sigma^{\mathrm{attosec}} \to \mathcal{B}_\sigma^{\mathrm{zeptosec}}.$$

These data satisfy

$$\Pi_\sigma^{\mathrm{zeptosec}} \circ \mathfrak{z}_\sigma^{\mathrm{zeptosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{attosec}}},$$

and the effective minimum value recovers the recorded atto-section functional:

$$\mathscr{A}_\sigma^{\mathrm{attosec}}(v) = \mathscr{U}_\sigma^{\mathrm{zeptosec}}(\mathfrak{z}_\sigma^{\mathrm{zeptosec}}(v)) = \min_{q \in \mathcal{F}_\sigma^{\mathrm{zeptosec}}(v)} \mathscr{U}_\sigma^{\mathrm{zeptosec}}(q) \qquad \text{for every } v \in \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

3. *A closed embedded exact zepto-section shell*

$$\mathcal{Z}_\sigma^{\mathrm{zeptosec}} \subset \mathcal{B}_\sigma^{\mathrm{zeptosec}}$$

satisfying

$$\mathcal{Z}_\sigma^{\mathrm{zeptosec}} = \left\{ q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}} : \begin{array}{l} \Pi_\sigma^{\mathrm{zeptosec}}(q) \in \mathcal{Z}_\sigma^{\mathrm{attosec}}, \\ \mathscr{U}_\sigma^{\mathrm{zeptosec}}(q) = \min\limits_{r \in \mathcal{F}_\sigma^{\mathrm{zeptosec}}(\Pi_\sigma^{\mathrm{zeptosec}}(q))} \mathscr{U}_\sigma^{\mathrm{zeptosec}}(r) \end{array} \right\},$$

and

$$\Pi_\sigma^{\mathrm{zeptosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{zeptosec}}} : \mathcal{Z}_\sigma^{\mathrm{zeptosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{attosec}}$$

is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as the fixed benchmark-relevant zepto-section package used on the active line.*

Definition 5 (Recorded zepto-section-generating substrate yocto-section dynamics)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. The recorded current yocto-section dynamics package $\mathfrak{C}^{\mathrm{yoctosec}}$ for sector σ consists of the following data.*

1. *A finite-dimensional Euclidean yocto-section state space $\mathcal{C}_\sigma^{\mathrm{yoctosec}}$, a compact convex yocto-section body*

$$\mathcal{B}_\sigma^{\mathrm{yoctosec}} \subset \mathcal{C}_\sigma^{\mathrm{yoctosec}}$$

with nonempty interior, and an affine surjection

$$\Pi_\sigma^{\mathrm{yoctosec}} : \mathcal{C}_\sigma^{\mathrm{yoctosec}} \to \mathcal{U}_\sigma^{\mathrm{zeptosec}}$$

satisfying

$$\Pi_\sigma^{\mathrm{yoctosec}}(\mathcal{B}_\sigma^{\mathrm{yoctosec}}) = \mathcal{B}_\sigma^{\mathrm{zeptosec}}.$$

For each $q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}}$, write

$$\mathcal{F}_\sigma^{\mathrm{yoctosec}}(q) \equiv \left(\Pi_\sigma^{\mathrm{yoctosec}}\right)^{-1}(q) \cap \mathcal{B}_\sigma^{\mathrm{yoctosec}}.$$

Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth yocto-section functional*

$$\mathscr{C}_\sigma^{\mathrm{yoctosec}} : \mathcal{C}_\sigma^{\mathrm{yoctosec}} \to [0,\infty)$$

such that, for every $q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}}$*, the restriction of* $\mathscr{C}_\sigma^{\mathrm{yoctosec}}$ *to* $\mathcal{F}_\sigma^{\mathrm{yoctosec}}(q)$ *is strictly convex and attains a unique minimizer. The resulting minimizer depends smoothly on* q*; write it as*

$$\mathfrak{y}_\sigma^{\mathrm{yoctosec}} : \mathcal{B}_\sigma^{\mathrm{zeptosec}} \to \mathcal{B}_\sigma^{\mathrm{yoctosec}}.$$

These data satisfy

$$\Pi_\sigma^{\mathrm{yoctosec}} \circ \mathfrak{y}_\sigma^{\mathrm{yoctosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{zeptosec}}},$$

and the effective minimum value recovers the recorded zepto-section functional:

$$\mathscr{U}_\sigma^{\mathrm{zeptosec}}(q) = \mathscr{C}_\sigma^{\mathrm{yoctosec}}\big(\mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)\big) = \min_{y \in \mathcal{F}_\sigma^{\mathrm{yoctosec}}(q)} \mathscr{C}_\sigma^{\mathrm{yoctosec}}(y) \qquad \text{for every } q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}}.$$

3. *A closed embedded exact yocto-section shell*

$$\mathcal{Z}_\sigma^{\mathrm{yoctosec}} \subset \mathcal{B}_\sigma^{\mathrm{yoctosec}}$$

satisfying

$$\mathcal{Z}_\sigma^{\mathrm{yoctosec}} = \left\{ y \in \mathcal{B}_\sigma^{\mathrm{yoctosec}} : \begin{array}{l} \Pi_\sigma^{\mathrm{yoctosec}}(y) \in \mathcal{Z}_\sigma^{\mathrm{zeptosec}}, \\ \mathscr{C}_\sigma^{\mathrm{yoctosec}}(y) = \min\limits_{x \in \mathcal{F}_\sigma^{\mathrm{yoctosec}}(\Pi_\sigma^{\mathrm{yoctosec}}(y))} \mathscr{C}_\sigma^{\mathrm{yoctosec}}(x) \end{array} \right\},$$

and

$$\Pi_\sigma^{\mathrm{yoctosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{yoctosec}}} : \mathcal{Z}_\sigma^{\mathrm{yoctosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{zeptosec}}$$

is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as a deeper substrate-side source for the current zepto-section package rather than as a replacement benchmark theory.*

Remark 1. *Definitions 4 and 5 fix the current frontier package. The present note does not spend its move on another deeper derivation below that frontier. It asks instead whether the full recorded yocto-section package carries benchmark-neutral ambient freedom that can be removed without changing the induced zepto-section package or the downstream chain that depends on it.*

3. Observer-Relevant Yocto-Section Minimizer-Image Core

Definition 6 (Observer-relevant yocto-section minimizer-image core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and a benchmark-faithful yocto-section package in the sense of Definition 5. The* observer-relevant yocto-section minimizer-image core $\mathfrak{C}^{\mathrm{yoc,img}}$ *for sector* σ *consists of:*

1. *the compact minimizer-image body*

$$\mathcal{B}_\sigma^{\mathrm{yoc,img}} \equiv \mathfrak{y}_\sigma^{\mathrm{yoctosec}}\big(\mathcal{B}_\sigma^{\mathrm{zeptosec}}\big) \subset \mathcal{B}_\sigma^{\mathrm{yoctosec}},$$

together with the restricted projection

$$\Pi_\sigma^{\mathrm{yoc,img}} \equiv \Pi_\sigma^{\mathrm{yoctosec}}\big|_{\mathcal{B}_\sigma^{\mathrm{yoc,img}}} : \mathcal{B}_\sigma^{\mathrm{yoc,img}} \to \mathcal{B}_\sigma^{\mathrm{zeptosec}};$$

2. *the restricted functional*

$$\mathscr{K}_\sigma^{\mathrm{yoc,img}} \equiv \mathscr{C}_\sigma^{\mathrm{yoctosec}}\big|_{\mathcal{B}_\sigma^{\mathrm{yoc,img}}};$$

3. *the exact-shell image*

$$\mathcal{Z}_\sigma^{\mathrm{yoc,img}} \equiv \mathfrak{y}_\sigma^{\mathrm{yoctosec}}(\mathcal{Z}_\sigma^{\mathrm{zeptosec}}) \subset \mathcal{Z}_\sigma^{\mathrm{yoctosec}};$$

4. *benchmark faithfulness in the sense of Definition 5(4).*

Remark 2. *Definition 6 is intentionally smaller than Definition 5. It keeps only the minimizer image over the recorded zepto-section body together with the restricted functional and the exact-shell image it already determines. The ambient yocto-section state space and off-image body directions are not taken as independent core data.*

4. Collapse of the Full Yocto-Section Package

Proposition 1 (The minimizer-image body carries the body-level content)**.** *For every benchmark-faithful yocto-section package, $\mathcal{B}_\sigma^{\mathrm{yoc,img}}$ is compact and the restricted projection*

$$\Pi_\sigma^{\mathrm{yoc,img}} : \mathcal{B}_\sigma^{\mathrm{yoc,img}} \to \mathcal{B}_\sigma^{\mathrm{zeptosec}}$$

is a homeomorphism with inverse $\mathfrak{y}_\sigma^{\mathrm{yoctosec}}$.

Proof. The body $\mathcal{B}_\sigma^{\mathrm{zeptosec}}$ is compact by Definition 4, and $\mathfrak{y}_\sigma^{\mathrm{yoctosec}}$ is continuous by Definition 5(2). Therefore

$$\mathcal{B}_\sigma^{\mathrm{yoc,img}} = \mathfrak{y}_\sigma^{\mathrm{yoctosec}}(\mathcal{B}_\sigma^{\mathrm{zeptosec}})$$

is compact. The identity

$$\Pi_\sigma^{\mathrm{yoctosec}} \circ \mathfrak{y}_\sigma^{\mathrm{yoctosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{zeptosec}}}$$

shows that $\Pi_\sigma^{\mathrm{yoc,img}}$ is surjective and that $\mathfrak{y}_\sigma^{\mathrm{yoctosec}}$ is a right inverse for it. If $y_1, y_2 \in \mathcal{B}_\sigma^{\mathrm{yoc,img}}$ satisfy

$$\Pi_\sigma^{\mathrm{yoc,img}}(y_1) = \Pi_\sigma^{\mathrm{yoc,img}}(y_2) = q,$$

then each y_i lies in the fiber $\mathcal{F}_\sigma^{\mathrm{yoctosec}}(q)$ and is the unique minimizer there, so $y_1 = y_2 = \mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)$. Thus $\Pi_\sigma^{\mathrm{yoc,img}}$ is bijective with inverse $\mathfrak{y}_\sigma^{\mathrm{yoctosec}}$. Since $\mathcal{B}_\sigma^{\mathrm{yoc,img}}$ is compact and $\mathcal{B}_\sigma^{\mathrm{zeptosec}}$ is Hausdorff, the continuous bijection $\Pi_\sigma^{\mathrm{yoc,img}}$ is a homeomorphism. □

Proposition 2 (The recorded zepto-section functional is carried by the image core)**.** *For every benchmark-faithful yocto-section package, the recorded zepto-section functional is recovered entirely from the restricted image-core functional:*

$$\mathscr{U}_\sigma^{\mathrm{zeptosec}}(q) = \mathscr{K}_\sigma^{\mathrm{yoc,img}}(\mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)) \qquad \textit{for every } q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}}.$$

Proof. By Definition 6(2),

$$\mathscr{K}_\sigma^{\mathrm{yoc,img}}(\mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)) = \mathscr{C}_\sigma^{\mathrm{yoctosec}}(\mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)).$$

Definition 5(2) then gives

$$\mathscr{C}_\sigma^{\mathrm{yoctosec}}(\mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)) = \mathscr{U}_\sigma^{\mathrm{zeptosec}}(q)$$

for every $q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}}$. □

Proposition 3 (The full exact yocto-section shell is forced by the image core)**.** *For every benchmark-faithful yocto-section package,*

$$\mathcal{Z}_\sigma^{\mathrm{yoctosec}} = \mathcal{Z}_\sigma^{\mathrm{yoc,img}} = \mathfrak{y}_\sigma^{\mathrm{yoctosec}}(\mathcal{Z}_\sigma^{\mathrm{zeptosec}}),$$

and the restricted projection

$$\Pi_\sigma^{\mathrm{yoc,img}}\big|_{\mathcal{Z}_\sigma^{\mathrm{yoc,img}}} : \mathcal{Z}_\sigma^{\mathrm{yoc,img}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{zeptosec}}$$

is a diffeomorphism.

Proof. If $y \in \mathcal{Z}_\sigma^{\mathrm{yoctosec}}$, then Definition 5(3) gives

$$q \equiv \Pi_\sigma^{\mathrm{yoctosec}}(y) \in \mathcal{Z}_\sigma^{\mathrm{zeptosec}},$$

and y minimizes $\mathscr{C}_\sigma^{\mathrm{yoctosec}}$ on the fiber $\mathcal{F}_\sigma^{\mathrm{yoctosec}}(q)$. By Definition 5(2), that minimizer is unique and equals $\mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)$. Hence

$$y = \mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q) \in \mathcal{Z}_\sigma^{\mathrm{yoc,img}},$$

so $\mathcal{Z}_\sigma^{\mathrm{yoctosec}} \subset \mathcal{Z}_\sigma^{\mathrm{yoc,img}}$.

Conversely, if $q \in \mathcal{Z}_\sigma^{\mathrm{zeptosec}}$, then $\mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)$ is by definition the unique minimizer of $\mathscr{C}_\sigma^{\mathrm{yoctosec}}$ on the fiber over q. Therefore $\mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q) \in \mathcal{Z}_\sigma^{\mathrm{yoctosec}}$, so $\mathcal{Z}_\sigma^{\mathrm{yoc,img}} \subset \mathcal{Z}_\sigma^{\mathrm{yoctosec}}$. This proves the equality of shells.

Finally,

$$\Pi_\sigma^{\mathrm{yoc,img}}\big|_{\mathcal{Z}_\sigma^{\mathrm{yoc,img}}} = \Pi_\sigma^{\mathrm{yoctosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{yoctosec}}},$$

and the latter is a diffeomorphism onto $\mathcal{Z}_\sigma^{\mathrm{zeptosec}}$ by Definition 5(3). □

Proposition 4 (The image core canonically reproduces the current zepto-section package)**.** *Fix a benchmark-faithful yocto-section package. Then the current zepto-section package of Definition 4 is exactly the benchmark-relevant package induced by the smaller image core. In particular:*

1. *the recorded zepto-section body and its body-level transport over that body are recovered by Proposition 1;*
2. *the recorded zepto-section functional on $\mathcal{B}_\sigma^{\mathrm{zeptosec}}$ is recovered by Proposition 2;*
3. *the recorded exact zepto-section shell is recovered by Proposition 3.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. Item (3) is Proposition 3. These statements are exactly the benchmark-relevant constituents of the current zepto-section package at current scope. □

Proposition 5 (All downstream recovery factors through the image core)**.** *Fix a benchmark-faithful yocto-section package. Then all downstream recovery of the current observer-relevant zepto-section minimizer-section core $\mathfrak{B}^{\mathrm{zep,sec}}$, the current observer-relevant zepto-section minimizer-image core*

$\mathfrak{B}^{\mathrm{zep,img}}$*, the current atto-section package* $\mathfrak{A}^{\mathrm{attosec}}$*, the current femto-section package* $\mathfrak{Z}^{\mathrm{femtosec}}$*, the current pico-section package* $\mathfrak{Y}^{\mathrm{picosec}}$*, the current observer-relevant pico-section minimizer-section core* $\mathfrak{Y}^{\mathrm{pic,sec}}$*, the current nano-section package* $\mathfrak{X}^{\mathrm{nanosec}}$*, the current micro-section package* $\mathfrak{W}^{\mathrm{microsec}}$*, the current sub-section package* $\mathfrak{V}^{\mathrm{subsec}}$*, and the previously recorded shell-transport consequences on the active line factors through* $\mathfrak{C}^{\mathrm{yoc,img}}$*.*

Proof. Definition 2(3) records that the active downstream chain depends on the recorded zepto-section package. Proposition 4 shows that this entire zepto-section package is already determined by the smaller image core. Therefore the downstream chain does not require the full ambient yocto-section package as additional independent data; it factors through $\mathfrak{C}^{\mathrm{yoc,img}}$. □

Proposition 6 (Ambient yocto-section directions are benchmark neutral at current scope)**.** *For every benchmark-faithful yocto-section package, the ambient yocto-section state space* $\mathcal{C}_\sigma^{\mathrm{yoctosec}}$ *and the off-image body directions*

$$\mathcal{B}_\sigma^{\mathrm{yoctosec}} \setminus \mathcal{B}_\sigma^{\mathrm{yoc,img}}$$

do not add independent benchmark-facing freedom beyond $\mathfrak{C}^{\mathrm{yoc,img}}$*.*

Proof. By Proposition 1, the full body-level content relevant to the recorded zepto-section package is already fixed by the minimizer-image body $\mathcal{B}_\sigma^{\mathrm{yoc,img}}$ and the restricted projection $\Pi_\sigma^{\mathrm{yoc,img}}$. By Proposition 2, the current zepto-section functional is already fixed by the restricted image-core functional $\mathscr{K}_\sigma^{\mathrm{yoc,img}}$. By Proposition 3, the full exact yocto-section shell is not additional data beyond the shell image; it is exactly that image. By Proposition 5, all downstream recovery of the current observer-relevant zepto-section minimizer-section core, the current observer-relevant zepto-section minimizer-image core, the current atto-section package, the current femto-section package, the current pico-section package, the current observer-relevant pico-section minimizer-section core, the current nano-section package, the current micro-section package, the current sub-section package, and the previously recorded shell-transport consequences factors through this smaller core. Finally, benchmark faithfulness is already part of Definition 6(4) and preserves the fixed outputs (1). Therefore nothing benchmark-facing at the present frontier requires the ambient yocto-section state-space presentation or body points outside the minimizer image as additional independent data. □

Proposition 7 (The benchmark package remains fixed)**.** *Every observer-relevant yocto-section minimizer-image core preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly* $g^\sharp$*, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definition 6(4). The present note only removes benchmark-neutral ambient yocto-section freedom from the live burden. It does not alter the benchmark exact-closure package. □

5. Main Theorem

Theorem 1 (Zepto-section-generating substrate yocto-section dynamics collapse to an observer-relevant yocto-section minimizer-image core)**.** *Fix the primitive continuation data of Definition 1, the recorded zepto-section downstream chain of Definition 2, the recorded zepto-section package of Definition 4, and a benchmark-faithful yocto-section package in the sense of Definition 5. Then, for each*

sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, the live benchmark-facing burden inside the current yocto-section frontier collapses from the full yocto-section package* $\mathfrak{C}^{\mathrm{yoctosec}}$ *to the smaller observer-relevant yocto-section minimizer-image core* $\mathfrak{C}^{\mathrm{yoc,img}}$*. More precisely:*

1. *the body-level content of the full yocto-section package is already carried by the minimizer-image body* $\mathcal{B}_\sigma^{\mathrm{yoc,img}}$*, the restricted projection* $\Pi_\sigma^{\mathrm{yoc,img}}$*, and the restricted image-core functional* $\mathscr{K}_\sigma^{\mathrm{yoc,img}}$ *by Propositions 1 and 2;*

2. *the full exact yocto-section shell is forced to equal the minimizer image of the recorded zepto-section shell by Proposition 3;*

3. *the current zepto-section package and therefore the current observer-relevant zepto-section minimizer-section core* $\mathfrak{B}^{\mathrm{zep,sec}}$*, the current observer-relevant zepto-section minimizer-image core* $\mathfrak{B}^{\mathrm{zep,img}}$*, the current atto-section package* $\mathfrak{A}^{\mathrm{attosec}}$*, the current femto-section package* $\mathfrak{Z}^{\mathrm{femtosec}}$*, the current pico-section package* $\mathfrak{Y}^{\mathrm{picosec}}$*, the current observer-relevant pico-section minimizer-section core* $\mathfrak{Y}^{\mathrm{pic,sec}}$*, the current nano-section package* $\mathfrak{X}^{\mathrm{nanosec}}$*, the current micro-section package* $\mathfrak{W}^{\mathrm{microsec}}$*, the current sub-section package* $\mathfrak{V}^{\mathrm{subsec}}$*, and the previously recorded downstream consequences all factor through* $\mathfrak{C}^{\mathrm{yoc,img}}$ *by Propositions 4 and 5;*

4. *ambient yocto-section directions outside that image core are benchmark neutral at current scope by Proposition 6;*

5. *the benchmark boundary remains fixed by Proposition 7;*

6. *consequently the remaining honest burden below the previous yocto-section frontier is no longer the full zepto-section-generating substrate yocto-section dynamics package* $\mathfrak{C}^{\mathrm{yoctosec}}$ *as such. What remains is to derive or justify the sharper image core* $\mathfrak{C}^{\mathrm{yoc,img}}$ *from still more primitive substrate structure, or else prove a still smaller theorem-level collapse strictly inside that image core.*

Proof. Item (1) is Propositions 1 and 2. Item (2) is Proposition 3. Item (3) is Propositions 4 and 5. Item (4) is Proposition 6. Item (5) is Proposition 7. Item (6) is the routing consequence of the previous items together with the active safeguard against counting another same-output deeper-origin relay as new front-line progress when the live benchmark-relevant datum is unchanged. □

Corollary 1 (What must be done next)**.** *After Theorem 1, the next honest benchmark-facing manuscript must either:*

1. *derive or justify the observer-relevant yocto-section minimizer-image core* $\mathfrak{C}^{\mathrm{yoc,img}}$ *from still more primitive substrate structure in a way that changes benchmark-facing status;*

2. *identify a genuinely smaller theorem-level observer-relevant datum strictly inside* $\mathfrak{C}^{\mathrm{yoc,img}}$ *and prove a sharper collapse to it;*

3. *do either of the previous two while preserving the same benchmark one-metric package and the same claim boundary.*

Another same-output deeper-origin relay that preserves this image core and therefore merely reproduces the same current zepto-section package $\mathfrak{B}^{\mathrm{zeptosec}}$*, the same current observer-relevant zepto-section minimizer-section core* $\mathfrak{B}^{\mathrm{zep,sec}}$*, the same current observer-relevant zepto-section minimizer-image core* $\mathfrak{B}^{\mathrm{zep,img}}$*, the same current atto-section package* $\mathfrak{A}^{\mathrm{attosec}}$*, the same current*

femto-section package $\mathfrak{Z}^{\mathrm{femtosec}}$*, the same current pico-section package* $\mathfrak{Y}^{\mathrm{picosec}}$*, the same current observer-relevant pico-section minimizer-section core* $\mathfrak{Y}^{\mathrm{pic,sec}}$*, the same current nano-section package* $\mathfrak{X}^{\mathrm{nanosec}}$*, and the same downstream micro-section and sub-section packages, another re-expansion back to the full yocto-section package as if its screened ambient directions were still live, another reopening of the already-screened current zepto-section package or the larger reconstructed presentations as if they were again the active burden, or any move that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger promotion language does not satisfy the present burden. If a quantum continuation is pursued, it matters benchmark-facingly only insofar as it derives this same image core, collapses it further, or closes a benchmark derivation gate in a genuinely new way.*

Proof. Theorem 1 shows that the full yocto-section package is no longer the minimal benchmark-facing datum at this stage. Therefore a subsequent manuscript must improve on the smaller image core rather than merely restating the larger package. □

6. Conclusion

The positive result of this note is a disciplined compression step inside the current yocto-section frontier. The previous active-primary theorem already removed the current zepto-section package as the primitive stopping point by deriving it from a deeper yocto-section package. The present theorem then takes the remaining honest internal route available there: it shows that the full yocto-section package is itself not the minimal benchmark-facing datum.

What survives as the live burden is the observer-relevant yocto-section minimizer-image core. That sharper datum consists of the minimizer-image compact body over the recorded zepto-section body, the restricted projection back to that body, the restricted yocto-section functional on that image, the exact-shell image of the recorded zepto-section shell, and benchmark faithfulness. The full exact yocto-section shell is forced to equal that image shell, and ambient yocto-section directions outside the minimizer image do not add independent benchmark-facing freedom once the induced zepto-section package is fixed.

This is the honest next burden after the present theorem. A new primary manuscript must derive or justify that sharper image core from still more primitive substrate structure, or else collapse it further, while preserving the same benchmark one-metric package and the same claim boundary. The current result does not yet complete a first-principles derivation of GR from substrate microphysics, but it does narrow the live derivational burden one level further on the active line.